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## Lab Report for EE202 Circuit Analysis

### *Laboratory 3*

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# 1 Introduction

This lab focuses on practical exploration of operational amplifier (op-amp) circuits, including non-inverting and inverting amplifiers, summing configurations, and gain analysis.

## 1.1 Preliminary Work

### 1.1.1 Preliminary Task 1

Gain: 10

1) Lab 3 Pre Work

$$R_1 = 1k \quad \frac{V_{in}}{R_1} = \frac{-V_{out}}{R_f}$$
$$R_f = 10k$$
$$\frac{V_{out}}{V_{in}} = \frac{-10k\Omega}{1k\Omega} = \boxed{-10 \text{ Gain}}$$

Figure 1: Inverting op-amp

### 1.1.2 Preliminary Task 2

$R_f = 10k$

2)  $V_{out} = V_{in} \left( 1 + \frac{R_f}{R_1} \right)$

$$R_1 = 10k\Omega$$
$$1 + \frac{R_f}{R_1} = 2$$
$$\boxed{R_f = 10k\Omega}$$

Figure 2: Inverting op-amp

### 1.1.3 Preliminary Task 3

$$3) V' = 0V$$

$$\frac{0-V_3}{R_3} + \frac{0-V_2}{R_2} + \frac{0-V_1}{R_1} = \frac{V_{out}-0}{R_f}$$

$$V_{out} = R_f \left( -\frac{V_3}{R_3} - \frac{V_2}{R_2} - \frac{V_1}{R_1} \right)$$

Figure 3: Summing Amplifier

### 1.1.4 Preliminary Task 4

Switch on A, Gain = 1; Switch on B, Gain = -1

|                                       |                                                              |
|---------------------------------------|--------------------------------------------------------------|
| 4) when SW on B                       | when SW on A                                                 |
| → inverting.                          | $\frac{V_{in} - V_{in}}{R_2} = \frac{V_{out} - V_{in}}{R_1}$ |
| $\text{Gain} = -\frac{R_1}{R_2} = -1$ | $V_{out} = V_{in}, \text{ Gain} = 1$                         |

Figure 4: Op-Amp Consisting of a Switch

### 1.1.5 Preliminary Task 5

5)

a) non inverting

$$V_{out} = \underbrace{(V_+ - V_-)}_3 \cdot A = 3A$$

$V_{out} = \underline{12}$  max saturation if  $A > 4$

b) Inverting

$$V_{out} = (V_+ - V_-)A = -3A$$

$$-3A < 12 \quad V_{out} = \underline{\underline{-12}}$$

c) negative feedback

$$V_{out} = A(3 - V_{out})$$

$$V_{out} = \frac{3A}{1-A} \approx \frac{3A}{A} \approx \underline{\underline{3V}}$$

$A \gg 1$

d) Positive feedback

$$V_{out} = A(V_{out} - 3) \quad \neq 3A$$

$$V_{out} = \frac{-3A}{1-A} \approx \underline{\underline{3V}}$$

Figure 5: Op-Amp

## 1.2 Lab Tasks

In this lab session, we completed 4 different tasks using the uA741 operational amplifier. Each task involved a different circuit setup to explore how the op-amp works.

### Important Note

Only the pre-lab work is included in this report. Due to time constraints, the lab task sections could not be completed.

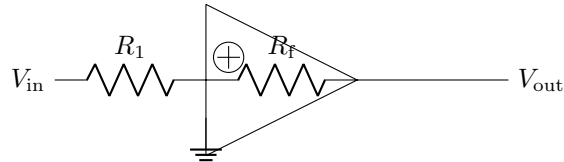


Figure 1: Op-amp